Week 5

Topic: Sliding Window

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932. Beautiful Array (Medium)

An array nums of length n is **beautiful** if:

- nums is a permutation of the integers in the range [1, n].
- For every 0 <= i < j < n, there is no index k with i < k < j where 2 * nums[k] == nums[i] + nums[j].

Given the integer n, return *any* ***beautiful*** *array* nums *of length* n. There will be at least one valid answer for the given n.

Example 1:

Input: n = 4

Output: [2,1,4,3]

For detailed descriptions and constraints, please refer to Leetcode 932.

995. Minimum Number of K Consecutive Bit Flips (Hard)

You are given a binary array nums and an integer k.

A **k-bit flip** is choosing a **subarray** of length k from nums and simultaneously changing every 0 in the subarray to 1, and every 1 in the subarray to 0.

Return *the minimum number of **k-bit flips** required so that there is no* 0 *in the array*. If it is not possible, return -1.

A **subarray** is a **contiguous** part of an array.

Example 1:

Input: nums = [0,1,0], k = 1

Output: 2

Explanation: Flip nums[0], then flip nums[2].

For detailed descriptions and constraints, please refer to Leetcode 995.

1499. Max Value of Equation (Hard)

You are given an array points containing the coordinates of points on a 2D plane, sorted by the x-values, where points[i] = [x_i, y_i] such that $x_i < x_j$ for all 1 <= i < j <= points.length. You are also given an integer k.

Return *the maximum value of the equation* $y_i + y_j + |x_i - x_j|$ where $|x_i - x_j| <= k$ and 1 <= i < j <= points.length.

It is guaranteed that there exists at least one pair of points that satisfy the constraint $|x_i - x_j| <= k$.

Example 1:

Input: points = [[1,3],[2,0],[5,10],[6,-10]], k = 1

Output: 4

Explanation: The first two points satisfy the condition $|x_i - x_j| <= 1$ and if we calculate the equation we get 3 + 0 + |1 - 2| = 4. Third and fourth points also satisfy the condition and give a value of 10 + -10 + |5 - 6| = 1.

No other pairs satisfy the condition, so we return the max of 4 and 1.

For detailed descriptions and constraints, please refer to Leetcode 1499.